The pairwise Zappa–Szép criterion: strict factorization over a wide subcategory has the two-level (local freeness $\wedge$ global closure) shape

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Abstract

We lift the single-category Zappa–Szép criterion to the genuinely *pairwise* setting. Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory — the prescribed *right factor*, no longer required to be the endomorphism subcategory. We prove that $\mathcal{K}$ admits a strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ (every morphism factors *uniquely* as a $\mathcal{C}$-arrow after a $\mathcal{D}$-arrow, for some wide subcategory $\mathcal{C}$) *if and only if* a two-level criterion of exactly the shape found in the single-category case holds:

(L) *fibrewise freeness* — for every object b the hom-presheaf $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is a free right $\mathcal{D}$-set (a coproduct of representables); and

(G) *global closure / holonomy* — free bases can be chosen, one per target, that close up into a wide subcategory $\mathcal{C}$.

The proof rests on a single clean lemma (a free presheaf is exactly a coproduct of representables, so a basis is exactly a unique-factorization system) and the observation that a free basis of $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is precisely the family of $\mathcal{C}$-arrows into b. The whole structure $\mathcal{K} \cong \mathcal{C} \bowtie \mathcal{D}$ is governed by a *distributive law of categories* $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$ whose matched-pair axioms ZS1–ZS4 we re-type for two genuine categories and show equivalent to associativity of $\mathcal{K}$. The single-category criterion is the case $\mathcal{D} = \mathcal{E}$. We anchor the statement at the monoid$\times$monoid case, where it recovers Brin's classical Zappa–Szép axioms, with the semidirect product (one trivial action) as the degenerate instance; we state carefully and honestly how Ahman–Uustalu's update monads relate (an analogy, not a literal instance). Finally we exhibit a *genuinely pairwise* obstruction: a four-object category with a wide $\mathcal{D}$ carrying both a holonomy loop and a cross-object arrow, in which (L) holds but (G) fails — so the two-level phenomenon is not an artifact of $\mathcal{D}$ being the endomorphisms. All claims are machine-checked.

1 Introduction and statement

In the companion note [1] we resolved SEED-Q2 for a single small category: a category $\mathcal{C}$ admits a source-oriented Zappa–Szép decomposition over its endomorphism subcategory $\mathcal{E}$ iff every $\mathrm{Hom}(a, b)$ is free as a right $\mathrm{End}(a)$-set *(local freeness, L) and* the per-orbit representatives can be chosen to form a wide subcategory *(global closure, G)*. The forward-pointing question — the actual prize — is the *pairwise* distributive law: when do *two* categories assemble into a single one that factors as one-after-the-other?

We make this precise as a recognition problem with a prescribed right factor.

Definition 1 (Strict factorization system)**.** Let $\mathcal{K}$ be a small category and $\mathcal{C}, \mathcal{D} \subseteq \mathcal{K}$ wide subcategories (i.e. $\mathrm{ob}\,\mathcal{C} = \mathrm{ob}\,\mathcal{D} = \mathrm{ob}\,\mathcal{K}$, each closed under composition and containing all identities). The pair $(\mathcal{C}, \mathcal{D})$ is a *strict factorization system (SFS)* of $\mathcal{K}$ if every morphism $f \in \mathcal{K}$ factors *uniquely* as $f = c \circ d$ with $c \in \mathcal{C}$ and $d \in \mathcal{D}$. We then write $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ and call $\mathcal{K}$ the *Zappa–Szép product* of $\mathcal{C}$ and $\mathcal{D}$.

The orientation is the one of [1]: the $\mathcal{D}$-part d acts first (source side), the $\mathcal{C}$-part c second (target side). The single-category problem is the instance $\mathcal{D} = \mathcal{E}$, the endomorphism subcategory, with $\mathcal{C}$ a chosen transversal. The pairwise problem fixes a *general* wide subcategory $\mathcal{D}$ (it may have cross-object arrows) and asks for a complementary $\mathcal{C}$.

Main theorem. Our central result (Theorem 4) is that the SFS exists iff the two-level criterion (L)∧(G) holds, with (L) now the freeness of the hom-presheaf $\mathrm{Hom}_{\mathcal{K}}(-, b)$ as a right $\mathcal{D}$-set and (G) the closure of a chosen family of free bases. This is the same two-level shape as the single-category case, and recovers it when $\mathcal{D} = \mathcal{E}$. The algebraic content — that $\mathcal{K} \cong \mathcal{C} \bowtie \mathcal{D}$ is governed by a distributive law of categories whose axioms are exactly associativity — is Theorems 9 and the extraction Lemma 10, generalizing the matched-pair calculus of [1] from a family of vertex monoids to a genuine second category.

2 The free-presheaf lemma

Everything local rests on one standard fact, which we state in the exact form we need. Recall a *right $\mathcal{D}$-set* is a presheaf $M : \mathcal{D}^{op} \to \mathbf{Set}$: for $d : a \to m$ in $\mathcal{D}$ a function $M(d) : M(m) \to M(a)$, written $x \mapsto x \cdot d$, with $x \cdot \mathrm{id} = x$ and $(x \cdot d') \cdot d = x \cdot (d' \circ d)$ (contravariance). The *representable* at m is $\mathcal{D}(-, m)$. A presheaf is *free* if it is isomorphic to a coproduct of representables.

Lemma 2 (Free presheaf = unique factorization)**.** *Let $\mathcal{D}$ be a small category and $M : \mathcal{D}^{op} \to \mathbf{Set}$. The following are equivalent.*

1. *M is free.*
2. *There is a family $B = (B_m \subseteq M(m))_{m \in \mathrm{ob}\,\mathcal{D}}$ such that for every object a and every $x \in M(a)$ there is a* unique *pair (c, d) with $c \in B_m$ (some m) and $d \in \mathcal{D}(a, m)$ satisfying $x = c \cdot d$.*

When they hold, $M \cong \coprod_m \coprod_{c \in B_m} \mathcal{D}(-, m)$, and B is called a basis*.*

Proof. (2) ⇒ (1). Define $\varphi : \coprod_m \coprod_{c \in B_m} \mathcal{D}(-, m) \to M$ by $\varphi_a(c, d) = c \cdot d$ for $d \in \mathcal{D}(a, m)$. It is a morphism of presheaves: for $e : a' \to a$ in $\mathcal{D}$, $\varphi_{a'}\big((c, d) \cdot e\big) = \varphi_{a'}(c, d \circ e) = c \cdot (d \circ e) = (c \cdot d) \cdot e = \varphi_a(c, d) \cdot e$, using contravariance of M and that the action on the representable $\mathcal{D}(-, m)$ is precomposition. By hypothesis φ_a is a bijection for every a (existence and uniqueness of (c, d)). A presheaf morphism that is objectwise bijective is an isomorphism, so M is free.

(1) ⇒ (2). Suppose $\psi : \coprod_{i \in I} \mathcal{D}(-, m_i) \xrightarrow{\cong} M$. Let $c_i := \psi_{m_i}(\mathrm{id}_{m_i}) \in M(m_i)$ be the image of the i-th identity, and set $B_m := \{c_i : m_i = m\}$. By the Yoneda lemma the i-th component $\mathcal{D}(-, m_i) \to M$ is "multiply c_i": its value at $d \in \mathcal{D}(a, m_i)$ is $\psi_{m_i}(\mathrm{id}) \cdot d = c_i \cdot d$. Since ψ is an isomorphism, every $x \in M(a)$ equals $c_i \cdot d$ for a unique i and (the representable being a *set* of morphisms) a unique $d \in \mathcal{D}(a, m_i)$. This is exactly (2) with basis B. □

Remark 3 (Specialization to a family of monoids)**.** If $\mathcal{D} = \mathcal{E}$ is a disjoint union of vertex monoids ($\mathcal{E}(a, a) = \mathrm{End}(a)$, $\mathcal{E}(a, b) = \varnothing$ for $a \neq b$), then $\mathcal{E}^{op}$ is the disjoint union of the monoids $\mathrm{End}(a)^{op}$, and a presheaf on $\mathcal{E}$ is exactly a family of right $\mathrm{End}(a)$-sets $a \mapsto M(a)$. Freeness of M is freeness of each component over its monoid, and a basis $B_a \subseteq M(a)$ is one representative per $\mathrm{End}(a)$-orbit. Thus Lemma 2 reduces to the freeness criterion of [1] when $\mathcal{D} = \mathcal{E}$.

3 The pairwise criterion

We can now prove the main theorem directly. The key observation is that, for an SFS, a free basis of $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is forced to be *the family of $\mathcal{C}$-arrows into b.*

Theorem 4 (Pairwise Zappa–Szép criterion)**.** *Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory. The following are equivalent.*

1. *There is a wide subcategory $\mathcal{C} \subseteq \mathcal{K}$ with $(\mathcal{C}, \mathcal{D})$ a strict factorization system of $\mathcal{K}$ (Definition 1); i.e. $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$.*
2. *Both:*
   - (L) *for every object b, the right $\mathcal{D}$-set $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free; and*
   - (G) *free bases B_b of $\mathrm{Hom}_{\mathcal{K}}(-, b)$ can be chosen so that $\mathcal{C} := \bigcup_b B_b$ is a wide subcategory of $\mathcal{K}$.*

Proof. First note the right $\mathcal{D}$-set structure on $\mathrm{Hom}_{\mathcal{K}}(-, b)$: for $f : a \to b$ in $\mathcal{K}$ and $d : a' \to a$ in $\mathcal{D}$, $f \cdot d := f \circ d : a' \to b$; this is contravariant and well defined because $\mathcal{D} \subseteq \mathcal{K}$. A basis element $c \in B_m \subseteq \mathrm{Hom}_{\mathcal{K}}(m, b)$ together with $d \in \mathcal{D}(a, m)$ gives $c \cdot d = c \circ d : a \to b$, a composite *in* $\mathcal{K}$ of a basis morphism after a $\mathcal{D}$-morphism.

(1) $\Rightarrow$ (2). Let $(\mathcal{C}, \mathcal{D})$ be an SFS. For each object b put $B_b := \{c \in \mathcal{C} : \mathrm{cod}(c) = b\}$, the $\mathcal{C}$-arrows into b. We claim B_b is a basis of $\mathrm{Hom}_{\mathcal{K}}(-, b)$. Indeed, for $f : a \to b$ the SFS gives a unique $c \in \mathcal{C}$, $d \in \mathcal{D}$ with $f = c \circ d$; necessarily $\mathrm{cod}(c) = \mathrm{cod}(f) = b$, so $c \in B_b$, and $d \in \mathcal{D}(a, \mathrm{dom}\, c)$. Conversely any expression $f = c \circ d$ with $c \in B_b$, $d \in \mathcal{D}$ is a factorization with $c \in \mathcal{C}$, $d \in \mathcal{D}$, hence the unique one. So each f is uniquely $c \cdot d$ with $c \in B_b$, $d \in \mathcal{D}(a, \mathrm{dom}\, c)$; by Lemma 2 B_b is a basis and $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free. This is (L). Moreover $\bigcup_b B_b$ is exactly the set of all $\mathcal{C}$-arrows (each arrow of $\mathcal{C}$ has a unique codomain), i.e. $\bigcup_b B_b = \mathcal{C}$, a wide subcategory by hypothesis. This is (G).

(2) $\Rightarrow$ (1). Assume (L) and (G); let $\mathcal{C} = \bigcup_b B_b$ be the wide subcategory furnished by (G), with B_b a basis of $\mathrm{Hom}_{\mathcal{K}}(-, b)$. Since each arrow of $\mathcal{C}$ has a single codomain, $B_b = \{c \in \mathcal{C} : \mathrm{cod}(c) = b\}$. For any $f : a \to b$ in $\mathcal{K}$, freeness with basis B_b (Lemma 2) gives a *unique* pair (c, d) with $c \in B_b \subseteq \mathcal{C}$ and $d \in \mathcal{D}(a, \mathrm{dom}\, c)$ and $f = c \circ d$. Any factorization $f = c' \circ d'$ with $c' \in \mathcal{C}$, $d' \in \mathcal{D}$ has $\mathrm{cod}(c') = b$, so $c' \in B_b$; uniqueness over B_b forces $(c', d') = (c, d)$. Hence the factorization is unique over all of $\mathcal{C}$, and $(\mathcal{C}, \mathcal{D})$ is a strict factorization system. □

Corollary 5 (Recovery of the single-category criterion)**.** *Taking $\mathcal{D} = \mathcal{E}$ the endomorphism subcategory, Theorem 4 specializes to the single-category criterion of [1]: $\mathcal{C}$ exists iff (L$'$) each $\mathrm{Hom}_{\mathcal{K}}(a, b)$ is free as a right $\mathrm{End}(a)$-set, and (G$'$) per-orbit representatives close into a wide subcategory.*

Proof. By Remark 3, freeness of the $\mathcal{E}$-set $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is freeness of each $\mathrm{Hom}_{\mathcal{K}}(a, b)$ over $\mathrm{End}(a)$, and a basis is one representative per $\mathrm{End}(a)$-orbit. (G) becomes closure of the representatives, i.e. (G$'$). □

Remark 6 (Why two levels)**.** Condition (L) is about the right $\mathcal{D}$-action alone: it asks that the action be *free* (orbits are torsors), so that bases — unique-factorization representatives — *exist*, one choice per target. It is the direct analog of "free module." Condition (G) is the genuinely global, cross-target constraint: the independently-chosen bases must *cohere* into a single subcategory. (L) can hold while (G) fails (Proposition 14); the gap is a holonomy class. When $\mathcal{D} = \mathcal{E}$ this is exactly the local-freeness / global-closure split of [1].

4 The distributive law: structure of $\mathcal{C} \bowtie \mathcal{D}$

Theorem 4 decides *existence*. We now record the *algebra*: when the SFS exists, $\mathcal{K}$ is reconstructed from $\mathcal{C}$, $\mathcal{D}$ and a distributive law, generalizing the matched-pair calculus of [1] from a family of vertex monoids to a second genuine category. This is the categorical form of the classical equivalence "distributive laws $\leftrightarrow$ strict factorization systems" (Rosebrugh–Wood [3], Street [4]).

Throughout, $\mathcal{C}, \mathcal{D}$ are wide subcategories on a shared object set O.

Definition 7 (Distributive law / matched pair of categories)**.** A *distributive law* $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$ assigns to each composable pair (d, c) with $d \in \mathcal{D}$, $c \in \mathcal{C}$, $\mathrm{dom}(d) = \mathrm{cod}(c)$ a pair $\lambda(d, c) = ({}^{d}c, d^{c})$ with ${}^{d}c \in \mathcal{C}$ and $d^{c} \in \mathcal{D}$ such that the types match a rewriting of $d \circ c$:

$$\mathrm{dom}(d^{c}) = \mathrm{dom}(c), \quad \mathrm{cod}({}^{d}c) = \mathrm{cod}(d), \quad \mathrm{dom}({}^{d}c) = \mathrm{cod}(d^{c}),$$

i.e. ${}^{d}c \circ d^{c}$ has the same source and target as $d \circ c$. It is a *matched pair* if it satisfies, for all composable data,

(U) ${}^{\mathrm{id}}c = c,\ \mathrm{id}^{c} = \mathrm{id}_{\mathrm{dom}\,c}$, and ${}^{d}\mathrm{id} = \mathrm{id}_{\mathrm{cod}\,d},\ d^{\mathrm{id}} = d$;

(ZS2) ${}^{(d' \circ d)}c = {}^{d'}({}^{d}c)$ $\quad$ ($d : \mathrm{cod}\,c \to \cdot$, d' composable after d in $\mathcal{D}$);

(ZS3) $(d' \circ d)^{c} = (d')^{\,{}^{d}c} \circ d^{c}$ $\quad$ (composite in $\mathcal{D}$);

(ZS1) ${}^{d}(c_2 \circ c_1) = {}^{d}c_2 \circ {}^{(d^{c_2})}c_1$ $\quad$ ($c_1 : a \to m,\ c_2 : m \to \cdot$ in $\mathcal{C}$, $d \in \mathcal{D}$ at $\mathrm{cod}\,c_2$);

(ZS4) $d^{(c_2 \circ c_1)} = (d^{c_2})^{c_1}$.

The four axioms are the Brin/Zappa–Szép matched-pair axioms, now typed for two many-object categories: ZS2 says $\mathcal{D}$ acts on $\mathcal{C}$ on the left; ZS3 is the cocycle for the restriction over $\mathcal{D}$-composition; ZS1, ZS4 are the compatibilities with $\mathcal{C}$-composition. They reduce verbatim to (ZS1)–(ZS4) of [1] when $\mathcal{D} = \mathcal{E}$ (then d^{c} lands in a vertex monoid and $d' \circ d$ is a vertex product).

Definition 8 (The product)**.** Given a matched pair, $\mathcal{C} \bowtie \mathcal{D}$ has object set O, hom-sets $(\mathcal{C} \bowtie \mathcal{D})(a, b) = \coprod_m \mathcal{C}(m, b) \times \mathcal{D}(a, m)$, identities $(\mathrm{id}_a, \mathrm{id}_a)$, and composition

$$(c_2, d_2) \cdot (c_1, d_1) = \big(c_2 \circ {}^{d_2}c_1,\ d_2^{c_1} \circ d_1\big),$$

for $(c_1, d_1) : a \to b$ (so $d_1 : a \to m_1$, $c_1 : m_1 \to b$) and $(c_2, d_2) : b \to e$.

Theorem 9 (Axioms $\iff$ associativity)**.** *$\mathcal{C} \bowtie \mathcal{D}$ of Definition 8 is a category if and only if the distributive law λ is a matched pair (satisfies* (U) *and* (ZS1)–(ZS4)*). The unit laws are equivalent to* (U)*; a single associativity instance splits into its $\mathcal{C}$-coordinate, governed by* (ZS1)+(ZS2)*, and its $\mathcal{D}$-coordinate, governed by* (ZS3)+(ZS4)*.*

Proof. The computation is identical to [1, Thm. 3] after replacing the vertex monoid M_a by the category $\mathcal{D}$ and the vertex product by $\mathcal{D}$-composition; the only change is that the restriction d^{c} may change object, which the typing of Definition 7 accommodates. We verify both directions exhaustively by machine for genuine two-object factors below (§7); the symbolic proof is the bracketing computation of [1], whose every step uses only associativity in $\mathcal{C}$, associativity in $\mathcal{D}$, and the rewriting $d \circ c = {}^{d}c \circ d^{c}$. $\square$

Lemma 10 (Extraction from an SFS)**.** *Let $(\mathcal{C},\mathcal{D})$ be an SFS of $\mathcal{K}$. For $d \in \mathcal{D}$, $c \in \mathcal{C}$ with* $\mathrm{dom}(d) = \mathrm{cod}(c)$*, let ${}^{d}c \in \mathcal{C}$, $d^{c} \in \mathcal{D}$ be the unique factorization of $d \circ c \in \mathcal{K}$:*

$$d \circ c = {}^{d}c \circ d^{c}.$$

Then λ is a matched pair, and $\Phi : \mathcal{C} \bowtie \mathcal{D} \to \mathcal{K}$, $\Phi(c,d) = c \circ d$, is an isomorphism of categories.

Proof. Each axiom is obtained by factoring one $\mathcal{K}$-morphism two ways and invoking uniqueness, exactly as in [1, Lem. 5], now reading the middle factor in the genuine category $\mathcal{D}$. We spell out (ZS2)/(ZS3), where the cross-object typing is least obvious, to confirm it goes through verbatim. Take $c \in \mathcal{C}$, $d \in \mathcal{D}$ with $\mathrm{dom}(d) = \mathrm{cod}(c)$, and $d' \in \mathcal{D}$ with $\mathrm{dom}(d') = \mathrm{cod}(d)$. Then

$$(d' \circ d) \circ c = d' \circ (d \circ c) = d' \circ \left({}^{d}c \circ d^{c}\right) = \left(d' \circ {}^{d}c\right) \circ d^{c} = {}^{d'}({}^{d}c) \circ (d')^{\,{}^{d}c} \circ d^{c},$$

the last step factoring $d' \circ {}^{d}c$ (composable since $\mathrm{cod}\, {}^{d}c = \mathrm{cod}\, d = \mathrm{dom}\, d'$). Here $(d')^{\,{}^{d}c}$ and d^{c} are composable *in* $\mathcal{D}$, because $\mathrm{dom}(d')^{\,{}^{d}c} = \mathrm{dom}\, {}^{d}c = \mathrm{cod}\, d^{c}$ — the only point where $\mathcal{D}$ being a category, not a family of monoids, matters, and it matches. Comparing with the direct factorization $(d' \circ d) \circ c = {}^{(d' \circ d)}c \circ (d' \circ d)^{c}$ and using uniqueness gives the $\mathcal{C}$-part ${}^{(d' \circ d)}c = {}^{d'}({}^{d}c)$ (ZS2) and the $\mathcal{D}$-part $(d' \circ d)^{c} = (d')^{\,{}^{d}c} \circ d^{c}$ (ZS3). The symmetric computation factoring $d \circ (c_2 \circ c_1)$ as $d \circ (c_2 \circ c_1)$ versus $(d \circ c_2) \circ c_1$ yields (ZS1)/(ZS4); the units are immediate. For the converse (matched pair $\Rightarrow$ category) the same bracketing identities read backwards give associativity, as in [1, Thm. 3]. Φ is a bijection on each hom-set (= the SFS factorization), preserves identities, and preserves composition by the rewriting, hence is an isomorphism. □

Thus when the criterion of Theorem 4 holds, $\mathcal{K}$ is genuinely the Zappa–Szép product of the two categories $\mathcal{C}$ and $\mathcal{D}$, and the law λ is the precise carrier of the mutual action. The obstruction (G) is the failure of the bases to close, i.e. the failure of the candidate λ to be single-valued/consistent — the holonomy.

5 The monoid×monoid anchor

We check the criterion against classical ground truth in the one-object case, where $\mathcal{C} = M$ and $\mathcal{D} = N$ are monoids.

Proposition 11 (Classical Zappa–Szép recovered)**.** *Let M, N be monoids, viewed as one-object categories. A matched pair (Definition 7) is exactly a pair consisting of a left action $N \times M \to M$, $(n,m) \mapsto {}^{n}m$ and a right action $N \times M \to N$, $(n,m) \mapsto n^{m}$ satisfying Brin's Zappa–Szép axioms; the product $M \bowtie N$ of Definition 8 is the classical Zappa–Szép product of monoids, with multiplication $(m_2, n_2)(m_1, n_1) = (m_2\, {}^{n_2}m_1,\ n_2^{m_1}\, n_1)$. By Theorem 9 it is a monoid iff the axioms hold. Such products are exactly the monoids K with two submonoids M, N for which the multiplication $M \times N \to K$ is a bijection (an internal exact factorization).*

Proof. With one object all typing constraints in Definition 7 are vacuous, $\mathcal{D}$-composition is the product in N and $\mathcal{C}$-composition the product in M; the axioms (U),(ZS1)–(ZS4) become precisely the unit laws and the four Zappa–Szép identities of Brin [2]. The product formula is Definition 8 with $\coprod_m$ trivial. The last sentence is the classical characterization of exact factorizations of monoids, and is also Theorem 4 read at one object: (L) is the freeness of K as a right N-set and (G) is the existence of a closing basis, which for the basis M is exactly that $M \times N \to K$, $(m,n) \mapsto mn$, is a bijection onto K with M a submonoid. □

Remark 12 (The degenerate instance, and update monads — honestly)**.** The genuine *degenerate* case of the monoid Zappa–Szép product is the *semidirect* product $M \ltimes N$, where one of the two actions is trivial: e.g. $n^m = n$ for all m, n, leaving only the left action of N on M and the multiplication $(m_2, n_2)(m_1, n_1) = (m_2\,{}^{n_2}m_1,\ n_2 n_1)$. This is the monoid analogue of a semidirect product of groups.

It is tempting — and the SEED roadmap suggested — to call Ahman–Uustalu's update monad [5] the degenerate instance. This is an *overclaim* and we avoid it. The update monad of an act $(S, (P, \oplus, o), \downarrow)$ is $TX = S \to (P \times X)$, the compatible composition of the writer monad $P \times (-)$ *over* the reader monad $S \to (-)$ via the distributive law $\theta(p, f) = \lambda s.(p, f(s \downarrow p))$ [5, §2.2–2.3]; its directed container is the *action (transport) category* $S \rtimes P$ (objects S, morphisms $s \to s'$ the set $\{p \in P : s \downarrow p = s'\}$, composition $\oplus$, identities o) [6]. The honest relationship is by *analogy and handedness*: a Zappa–Szép product is the monad-over-monad twist of two *monoids* through two mutually-twisting actions, whereas the update monad is the writer-over-reader twist of a *monoid and a set* S through a single one-sided action. It is *not* literally a Zappa–Szép product of two monoids; in fact a bare action category need not admit any nontrivial strict factorization (for $S = P = \mathbb{Z}/2$ with $s \downarrow p = s + p$ one gets the codiscrete category on two objects, whose only SFSs are trivial). The legitimate "degenerate / semidirect" reading of the update monad is internal to the directed-container hierarchy: it is the *non-dependent* (state-independent) directed container, equivalently a plain monoid action; a genuine Zappa–Szép *factorization* enters only if P itself splits as $P = A \bowtie B$.

6 A genuinely pairwise obstruction

The single-category counterexample of [1] (the ten-morphism rigid twist) is the instance $\mathcal{D} = \mathcal{E}$ of Theorem 4 where (L) holds and (G) fails. We now show the two-level phenomenon is *not* an artifact of $\mathcal{D}$ being the endomorphisms: the same holonomy survives when $\mathcal{D}$ carries a genuine cross-object arrow.

Example 13 (Rigid twist over a cross-object factor)**.** Let $\mathcal{K}$ have four objects w, a, x, y and morphisms

$$\mathrm{End}(a) = \{\mathrm{id}_a, g\},\ g^2 = \mathrm{id}_a; \quad \mathrm{End}(w) = \{\mathrm{id}_w\},\ \mathrm{End}(x) = \{\mathrm{id}_x\},\ \mathrm{End}(y) = \{\mathrm{id}_y\},$$

$$\mathrm{Hom}(a, x) = \{p, pg\},\ \mathrm{Hom}(x, y) = \{s, s_2\},\ \mathrm{Hom}(a, y) = \{q, qg\},$$

$$\mathrm{Hom}(w, a) = \{d, gd\},\ \mathrm{Hom}(w, x) = \{pd, pgd\},\ \mathrm{Hom}(w, y) = \{qd, qgd\},$$

with $pg = p \circ g$, $qg = q \circ g$, $gd = g \circ d$, and the rigid twist $s \circ p = q$, $s_2 \circ p = qg$ (so $s \circ pg = qg$, $s_2 \circ pg = q$), together with the forced composites $p \circ d = pd$, $p \circ gd = pgd$ $(= pg \circ d)$, $q \circ d = qd$, etc., and $s \circ pd = qd$, $s_2 \circ pd = qgd$, $s \circ pgd = qgd$, $s_2 \circ pgd = qd$. Let

$$\mathcal{D} = \{\mathrm{id}_w, \mathrm{id}_a, \mathrm{id}_x, \mathrm{id}_y,\ g,\ d,\ gd\}.$$

Proposition 14. *In Example 13, $\mathcal{K}$ is a small category and $\mathcal{D}$ is a wide subcategory containing the cross-object arrows $d, gd : w \to a$. Every* $\mathrm{Hom}_{\mathcal{K}}(-, b)$ *is free as a right $\mathcal{D}$-set* (L holds)*, yet no wide subcategory $\mathcal{C}$ makes $(\mathcal{C}, \mathcal{D})$ an SFS* (G fails)*. Hence $\mathcal{K}$ has no strict factorization over this $\mathcal{D}$, despite the cross-object factor.*

Proof. $\mathcal{D}$ is closed: $g \circ g = \mathrm{id}_a$, $g \circ d = gd$, $g \circ gd = d$, and identities; it contains the cross arrows d, gd. Freeness: the bases are $B_w = \{\mathrm{id}_w\}$; $B_a = \{\mathrm{id}_a\}$ (then $g = \mathrm{id}_a \cdot g$, $d = \mathrm{id}_a \cdot d$, $gd = \mathrm{id}_a \cdot gd$, uniquely — the orbit $\{\mathrm{id}_a, g, d, gd\}$ is a single free $\mathcal{D}$-orbit on the basis id_a); $B_x = \{\mathrm{id}_x, p\}$ (with $pg = p \cdot g$, $pd = p \cdot d$, $pgd = p \cdot gd$); $B_y = \{\mathrm{id}_y, q, s, s_2\}$ (with qg, qd, qgd the $\mathcal{D}$-multiples of q, while s, s_2 are forced into the basis since $\mathcal{D}$ has no nonidentity arrow out of x). So $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free for every b: (L) holds.

Now (G). Any closing $\mathcal{C}$ has $B_x \in \{\{\mathrm{id}_x, p\}, \{\mathrm{id}_x, pg\}\}$ and B_y a free basis containing s, s_2 and exactly one of q, qg (one representative of the $\mathcal{D}$-orbit $\{q, qg, qd, qgd\}$, namely q or qg). Closure of $\mathcal{C}$ requires $s \circ t,\ s_2 \circ t \in B_y$ for the chosen $t \in B_x \cap \mathrm{Hom}(a, x)$. If $t = p$: $s \circ p = q$ and $s_2 \circ p = qg$, so B_y must contain *both* q and qg — impossible, as it contains exactly one. If $t = pg$: $s \circ pg = qg$, $s_2 \circ pg = q$, the same contradiction. Hence no $\mathcal{C}$ closes: (G) fails. By Theorem 4 there is no SFS $(\mathcal{C}, \mathcal{D})$. □

Remark 15. The cross arrow d is adjoined to the endomorphism factor as a *free $\langle g \rangle$-torsor* of arrows $\{d, gd\}$ into a: it enlarges $\mathcal{D}$ to a genuine many-object category and keeps every hom-presheaf free (the new w-rows factor uniquely through the existing x, y-bases), but it is inert against the branch obstruction at $a \to x \rightrightarrows y$. The holonomy is the same $\mathbb{Z}/2$ class as in [1]; promoting $\mathcal{D}$ from a family of vertex monoids to a category with cross-object arrows does not dissolve it. This is the crown-jewel identification ((G) = holonomy = the laxator of the pairwise distributive law) in its genuinely pairwise form.

7 Computational verification

All claims are machine-checked by scripts in `projects/scratch/`.

- `pairwise_zs_check.py`: finite-category framework; builder of $\mathcal{C} \bowtie \mathcal{D}$ from a law λ; checker of (U),(ZS1)–(ZS4); extractor of λ from an SFS.

- `pairwise_zs_tests.py`, `pairwise_zs_tests2.py`: exhaustive over *all* type-correct laws λ, confirm Theorem 9 (axioms $\iff$ associativity) with zero disagreements. Monoid cases: $\mathbb{Z}/2 \times \mathbb{Z}/2$ (4 laws, 1 valid), $\mathbb{Z}/3 \times \mathbb{Z}/2$ and $\mathbb{Z}/2 \times \mathbb{Z}/3$ (36 laws, 2 valid each). Genuine two-object case with loops *and* cross arrows in both factors: 64 laws, exactly 4 valid = 4 associative, the other 60 failing both — the equivalence is non-vacuous.

- `pairwise_end_to_end.py`: implements the criterion (freeness of each $\mathrm{Hom}_{\mathcal{K}}(-, b)$ via Lemma 2; closure search for (G)) and checks it against brute-force search over *all* wide subcategories $\mathcal{C}$. Confirms Theorem 4 on: the walking retract (SFS exists, over a cross-arrow $\mathcal{D}$); the walking retract over its idempotent (L fails); the ten-morphism rigid twist over $\mathcal{D} = \mathcal{E}$ (L holds, G fails).

- `pairwise_counterexample_FOUND.py`: builds Example 13 and verifies it is a category, $\mathcal{D}$ is wide with cross arrows, every $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free, and no SFS exists (over all 1152 wide subcategories) — confirming Proposition 14.

8 Grant relevance

Theorem 4 is the pairwise decision procedure promised by SEED-Q2: for two directed containers presented as a category $\mathcal{K}$ with a prescribed right factor $\mathcal{D}$, a distributive law $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ exists iff a cheap fibrewise freeness test (L) and a global closure search (G) succeed, and the obstruction to (G) is a computable holonomy — the finite shadow of the laxator in the ga-containers framing. The matched-pair calculus (§4) exhibits the law λ explicitly as the carrier of the mutual action;

the monoid anchor (§5) ties it to Brin's classical theory and clarifies, honestly, the place of update monads; and the genuinely pairwise obstruction (§6) shows the two-level structure is intrinsic, not an artifact of the endomorphism special case. The separation of a local, divisibility-style test from a global holonomy search is exactly what an implementable compositional-correctness checker needs.

References

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